Week 10

Keel Bone Chicken Project

08/10/26

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This Week

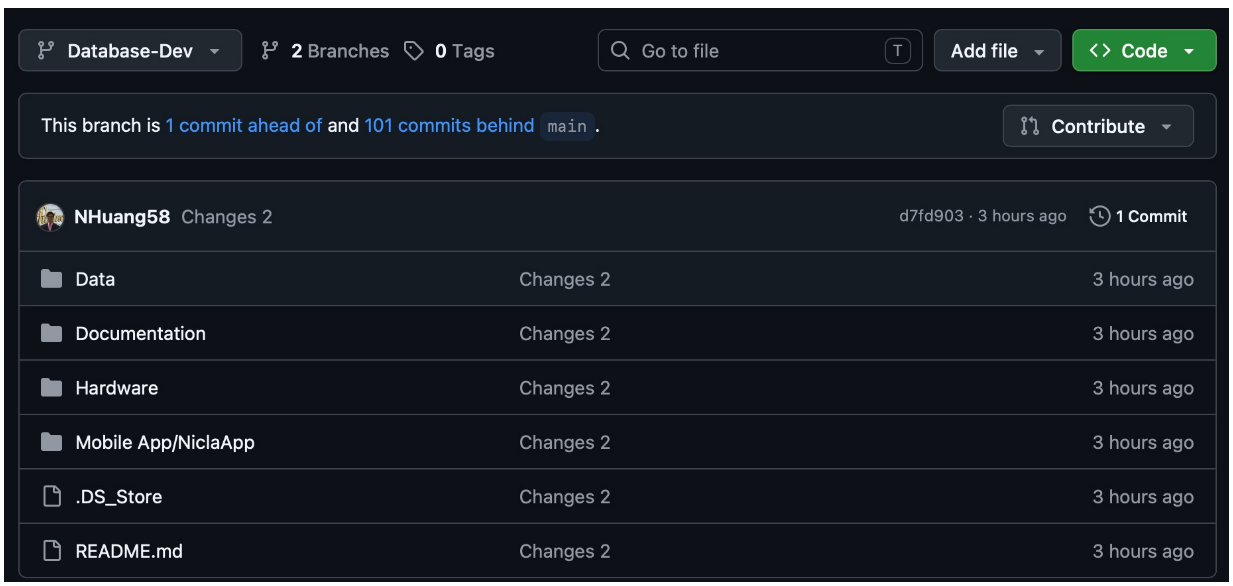
Weekly Progress

- Web Logger Updates
  - Logger complete
- PCB Updates
  - Working on placement file updates
- Documentation Updates
  - GitHub Documentation Created
  - GitHub Files Transferred
  - Ownership Transferred
- Paper Updates
  - Paper Acceptance Date Changed
- Next Week
  - Plans for next week
  - Overall Thoughts

Web Logger Updates

Logger Complete

- All features in Web Logger
- Logger in separate branch of GitHub
- Changed database to SQLite3 database
  - Everything functional



PCB Updates

Working on placement file updates

- Noticed discrepancy with component placements and drill file
- Contacted DL-Engineering; in the process of fixing
- Will have to go to procurement office for new quote + purchasing

Documentation Updates

GitHub Documentation Created

- GitHub Documentation Created to maintain consistency moving forward
- Mostly in .MD files

Monitoring

This document serves as the standard operating procedure and technical manual for deploying, operating, and managing data from the custom miniaturized Printed Circuit Board (PCB) designed to monitor keel bone dynamics in poultry.

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1. Hardware Specifications and Setup

1.1 Physical Components

The biomedical sensor node consists of a custom four-layer PCB integrated into an animal-wearable form factor. Key components include:

- **Microcontroller:** Nordic Semiconductor nRF52840 SoC (supporting ARM Cortex-M4F and Bluetooth Low Energy 5.0 protocols).
- **Inertial Measurement Unit (IMU):** TDK InvenSense ICM-20948 9-axis MEMS sensor (3-axis gyroscope, 3-axis accelerometer, and 3-axis compass).
- **Power Supply:** Onboard rechargeable Lithium-Polymer (Li-Po) battery circuit regulated by a Microchip MCP73831 charge management controller.
- **Storage:** External SPI flash memory for high-fidelity data logging when wireless telemetry is obstructed.

1.2 Encapsulation and Biocompatibility

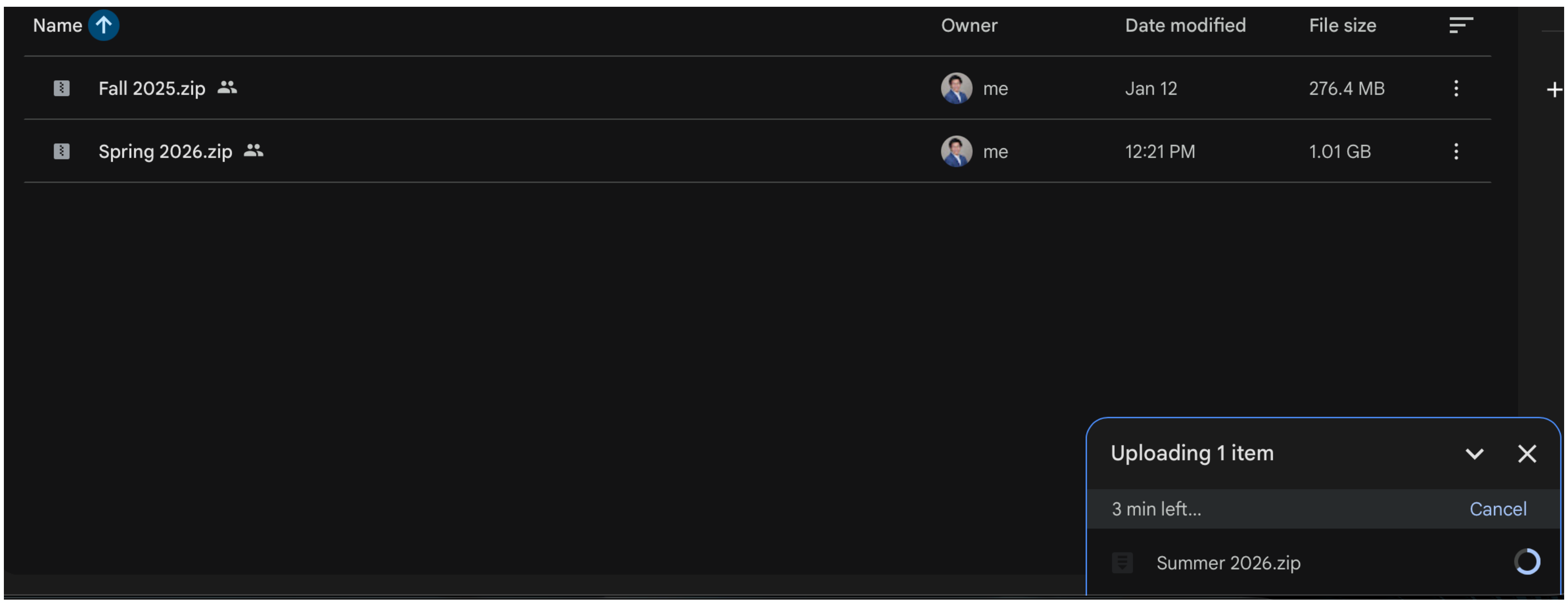
To protect the electronics from moisture, bodily fluids, and mechanical stress, the hardware must be encapsulated before deployment:

1. Ensure the battery is securely attached and the board has been programmed and post-assembly tested.
2. Verify that the total package weight does not exceed the target threshold for the specific avian subject to avoid altering natural gait

Documentation Updates

GitHub Files Transferred

- Files Transferred over into Google Drive:
- https://drive.google.com/drive/folders/1Ul3LAQrhO1t5J_EVqvNEP-hPAe9nVNyN



Documentation Updates

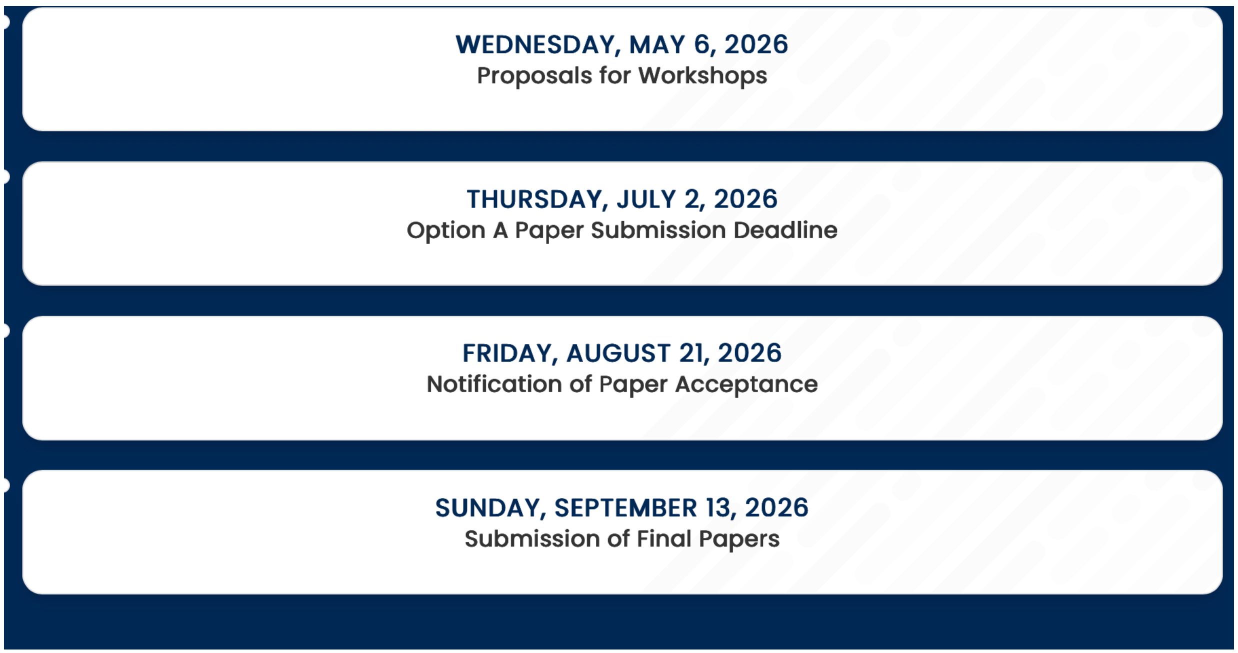
Ownership Transferred

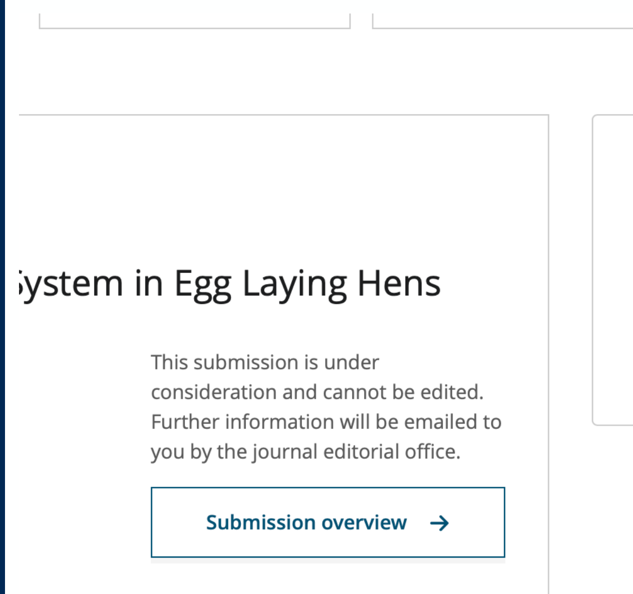
- Transferred ownership of repository over to Nolan
- Should take a few days

Paper Updates

Paper Acceptance Date Changed

- Now notification of Acceptance will be on 8/21
- No updates on author portal





Next Week

Plans for next week

- Get PCB fully ordered and ensure payment goes through
- Wrap up documentation for next developer

Next Week

Thoughts

- Very useful interdisciplinary project where I was able to practice a lot of skills
  - Firmware Development
  - PCB Design
  - Sensor Testing
  - Communication with other parties
- Lot of potential for the project; can’t wait to see where it goes

Thank You

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